

5 IN 5

In this series of 5 videos, IBM research

scientists speak about the future of cognitive computing pertaining to the five senses and how devices around these senses will revolutionise the way we use technology. Probable applications of such devices are immense. <http://dgit.in/XNAM3Y>

UNDERSTANDING LYTRO

The refo-cussable images coming

out of the Lytro camera have impressed us. This article goes behind the aesthetics of light field photography and speaks about the pros and cons of the technology. <http://dgit.in/T9appj>

8K: THE ULTIMATE IN TV SCREENS?

BBC Click team visits Japan, where its leading tech brands are working together on an 8,000 pixel-wide display which they call the Super Hi-vision. According to the presenter, this TV screen is even more immersive than traditional 3D. Consumer 8K TVs are still 7 years away though. <http://dgit.in/WuVlXI>

SURREAL SELF-PORTRAITS

A photograph of you carrying yourself may make you wonder what we have been smoking, but this tutorial teaches you just how to do that! <http://dgit.in/Tvkugg>

advanced civilization or individual would seek to simulate the unfolding of its own universe, how life came to be in that universe, and how the simulator itself came to be, perhaps even nested within another simulation.

So suppose we suggest our own axiom as follows — “the simulation seeks to maximize the richness of the environment in which minds might develop, interact and probe the constraints of their own simulation while at the same time minimizing the amount of suffering of the unfortunate among those that arise in that simulation.” While the simulators may have had good intentions and respect for life, it is impossible to deny that there is and has been non-negligible suffering in our simulation. Is an axiom such as this compatible with what we can now observe?

One way to view our present simulation in this light is to imagine it set in motion with incomplete knowledge of what would unfold when left to its own devices. Perhaps the simulators cannot, or have chosen not to observe or manipulate the simulation. The original simulators might have long since expired or otherwise become detached from the simulation, having exhausted their own energy supply in an open universe, or having collapsed to a singularity in a closed one, somehow managing to manipulate matter on a scale grand enough to accelerate the simulation beyond collapse.

YOU ARE THE ONE

Another way to examine the simulation is through the calculus of extremes. An appeal to the extreme viewpoint of the solipsist – the belief that the only sure thing in the universe is your own mind — would address the problem of minimization of suffering because it can simply be denied. The only discomfort in existence would be one's own.



Why then might a simulator even bother with simulating the environment if the most interesting thing to simulate is a mind?

If in fact many solipsistic minds are simulated, each in their own isolated world, the computational complexity of the task would grow enormously, making it desirable to at some point to overlap the environments. We might then ask, where in this kind of arrangement do we fall?

One question a simulation might attempt to answer is what is the optimal number of minds to apply to the major problems of the universe. I.e. how to maximally extend and allocate its finite resources. In other words, do you want a just a few large, hyper-intelligent minds attacking the problem, or rather lots of smaller minds?

To have faith that every water molecule in the ocean is being perfectly simulated at this very moment, in the absence of observation, may be a testament to just how easy it is for our universe to simulate reality. The number of permissible states for an ocean's atoms would seem near infinite compared to the those for the three-pound mass in our heads, but which is really easier to simulate? The movements of the ocean can easily be captured with principles and even laws which could vastly simplify the task. Per-

haps one day we will find a way to test this kind of hypothesis.

The idea of setting up some kind of “glitch” detector to look for inconsistencies in the simulation has been raised, but it may be difficult to characterize exactly what a glitch would be. People have experiences all the time in the normal operation of their own brains which could just as well be interpreted as as glitch, but typically we do not define them as such. Whereas a seemingly vivid apparition of a long since loved one in the hallway at night may be experienced fully from the incidental stimulus of a few errant sounds, another person would no doubt interpret the situation in an entirely different way.

The question of the degree to which simulations and minds may overlap is a question for another day. We are just beginning to write the experiments that create the conclusions we might draw, in the manner of the particle physicist. Solipsistic inspection is certainly useful as a tool for thought experiments, but as a social strategy it can only be a failure and is largely discounted as such. The viewpoint does offer one subtle abode — there is evidence that other people age and in fact die, but for you, there is only evidence that you get older.